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SMART ATTENDANCE SYSTEM USING FACE RECOGNITION AND MACHINE LEARNING

Divyanshu Giri*1, Aman Rana*2, Nikhil*3, Yug Pratap Singh*4, Abhishek Mishra*5, Mr.Ashutosh Kumar*6

Department Of Computer Science & Engineering, HMR Institute Of Technology And Management, New Delhi, India

Assistant Professor, Department Of Computer Engineering, HMR Institute Of Technology And Management, New Delhi, India.

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ABSTRACT

In both business and educational settings, attendance management is a required procedure, yet traditional approaches are still ineffective. Conventional methods, including roll calls or paper-based signing sheets, are time-consuming, labour-intensive, and very vulnerable to fraudulent activities like proxy attendance. This study describes the creation and deployment of a smart attendance system that uses facial recognition technology to automate the recording process. The K-Nearest Neighbours (KNN) technique is used in the proposed system to classify facial features that are captured by a camera and analysed using OpenCV. The technology automatically records attendance in a CSV file and offers audible confirmation when it detects a registered face. Additionally, administrators can check daily attendance records in real-time through a web interface built on Streamlit. This solution minimizes human error, saves valuable class time, and ensures a secure, tamper-proof record of attendance.

Keywords: Face Recognition, Machine Learning, OpenCV, K-Nearest Neighbors, Streamlit, Automated Attendance.

I. INTRODUCTION

Monitoring attendance is an essential part of assessing the regularity and performance of employees in business organisations and students in educational institutions. Accurate, reliable, and secure record-keeping is necessary for effective management. In the past, vocal roll calls and the distribution of signature forms were used to manually record attendance. These manual techniques often lead to problems including human calculation errors, redundant data, and lost teaching time. Furthermore, the integrity of the record is compromised by the issue of "buddy punching," in which one person records attendance for a peer who is not present.

Automation has become essential for improving organisational processes as technology develops at a rapid pace. The verification procedure is streamlined by biometric technology, particularly Face Recognition, which provide an effective and non-intrusive digital platform. Face recognition uses computer vision to identify people from a distance, in contrast to fingerprint scanners, which necessitate physical touch and hygienic considerations.

The goal of this project is to develop and deploy a Python-based smart attendance application that simplifies the entire attendance process. The system uses KNN for recognition and Haar Cascade classifiers for face detection in an effort to improve the user experience. It is suitable for workplaces, seminars, and classrooms when prompt and precise attendance is needed without the need for human interaction.

II. LITERATURE REVIEW

The development of information technology has significantly transformed traditional management techniques into tech-based, automated solutions. Several models have been proposed by numerous scholars to improve the accuracy and efficiency of attendance tasks.

Barcode technology and Radio Frequency Identification (RFID) were major components of early automated systems. These systems decreased human paperwork, but because ID cards could be readily switched between users, they were inadequate for confirming a person's actual presence. This demonstrated how biometric

verification is necessary to ensure data accuracy. Fingerprint recognition systems became the subject of later research. These systems have issues with scalability and hygiene, especially in heavy-traffic areas, despite their great accuracy. During peak hours, the wait time to place a finger on a sensor turned out to be a bottleneck.The emphasis now is on facial recognition due to recent advances in computer vision. Viola and Jones (2001) developed a foundation for real-time object detection using Haar Cascades, a technique that is still widely used for face detection because of its speed. In order to link face traits to identify, contemporary applications incorporate machine learning methods such as K-Nearest Neighbours (KNN) or Deep Learning (CNNs).Nevertheless, a lot of the commercial systems that are now in use are costly or demand a lot of hardware. In order to overcome these limitations, this project develops a lightweight, PC-based solution that integrates OpenCV for image processing and Streamlit for an approachable, user-friendly dashboard, successfully combining essential elements like face registration, real-time recognition, and CSV-based reporting into a single, coherent system.

III. PROBLEM STATEMENT

Manually managing attendance is a time-consuming and error-prone process that often results in inefficiencies and a lack of cooperation in both corporate and educational settings. Conducting verbal roll calls or distributing attendance papers frequently wastes important teaching time in colleges and universities. These functions have traditionally been performed using physical registers, which are vulnerable to loss, destruction, and unauthorised manipulation. "Buddy Punching" or proxy attendance, in which one student notes the presence of an absent peer, is a major problem with the existing manual approach. This makes it challenging for administrators to monitor real involvement and jeopardises the integrity of academic records. Additionally, manually entering data into spreadsheets for monthly reporting takes a lot of effort and increases the possibility of human error, such as listing a current student as absent. Current biometric systems, like fingerprint scanners, frequently need costly hardware installation and raise hygienic issues because of direct touch. Therefore, in order to automate the identification, logging, and reporting operations, a computerised and contactless Smart Attendance Application is required. This solution must ensure data accuracy, reduce paperwork, use facial recognition to avoid proxy attendance, and provide a single platform for efficiently managing daily records in real-time.

IV. OBJECTIVES OF THE STUDY

This study's primary objective is to develop and deploy a Python-based smart attendance system that simplifies and makes it easier to track and record attendance.

The following are the precise goals:

1) To automate user identification: To ensure contactless verification, a computer vision system that uses OpenCV and Haar Cascade Classifiers to detect and identify faces will take the role of manual roll calls.

2)To guarantee precise identification: To employ the K-Nearest Neighbours (KNN) machine learning technique to accurately categorise facial embeddings in order to differentiate between registered users.

3) To avoid fraud and redundant data: To provide a safe enrolment procedure in which a user's face is photographed 100 times to produce a reliable dataset, avoiding proxy attendance and false positives.

4) In order to give immediate feedback: Using text-to-speech technology (win32com), audio feedback will be integrated to communicate to the user that their attendance has been correctly recorded without requiring them to view the screen.

5) Data administration can be streamlined by automatically creating and updating daily CSV files (such as Attendance_27-11-2025.csv), which would remove the need for human data entry.

6) To make a dashboard that is accessible: to use Streamlit to create a web-based interface that enables

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6) To make a dashboard that is accessible: to use Streamlit to create a web-based interface that enables

managers to view, track, and evaluate attendance records in real-time.

By achieving these objectives, the study aims to provide a centralised, efficient, and user-friendly digital solution that addresses the shortcomings of traditional manual attendance.

V. METHODOLOGY

Requirement analysis, data collection, model training, and system integration are the four main stages of the structured research methodology used to build the Face Recognition Attendance System. This approach aims to provide seamless data management and biometric recognition accuracy.

Analysis of Requirements Based on the drawbacks of manual attendance, the functional needs were determined during this phase.

Among the essential prerequisites are:

1)The capacity to record and retain each user's distinct facial traits is known as biometric data collection.
2)Machine Learning Integration: Recognising registered users from live video using a classification system.
3)In real time Feedback: After successful attendance marking, the user receives an audio confirmation.
4)Dashboard Visualisation: An online tool that allows managers to dynamically view attendance records.

Technologies and Tools for Development The following Python-based stack was used in the system's development:

1)Computer Vision: To access the webcam and carry out image processing operations like converting frames to greyscale (cv2.cvtColor) and identifying faces using Haar Cascades, OpenCV (Open Source Computer Vision Library) was utilised.

2)Machine Learning: The K-Nearest Neighbours (KNN) technique for classification was implemented using Scikit-learn.

3)Data handling: Daily attendance records were managed using Pandas and CSV, while facial data (faces_data.pkl) and labels (names.pkl) were serialised using Pickle.

4)User Interface: The online dashboard was created using Streamlit, and Streamlit Autorefresh was used to guarantee that the data table updated automatically without requiring page reloads.

VI. SYSTEM DESIGN

The Face Recognition Project's architecture, components, data flow, and interactions between its various modules are all specified in the System Design.

Architecture of the System The program has a modular architecture, with separate scripts handling data gathering, processing, and presentation**.**

1) Input Layer: The video feed is captured by the actual webcam**.**

2) Layer of Processing:

. New user enrolment is handled by add_faces.py.

. The live recognition and logic (name prediction and time marking) are handled by test.py.

3) Layer of Storage:

. Pickle Files: Keep the training data locally (names.pkl, faces_data.pkl).

. Keep the daily attendance records in CSV files, such as Attendance_27-11-2025.csv.

4) Presentation Layer: After reading the CSV, the Streamlit dashboard (app.py) displays the information to the administrator.

System Modules

Three main modules make up the application:

1. Add_faces.py, the Face Registration Module The database is filled in by this module**.**

. Both the webcam and the Haar Cascade classifier (haarcascade_frontalface_default.xml) are initialised**.**

. The user is prompted to input their name**.**

. The user's face is captured in 100 frames. Before being added to the faces_data list, each captured face is cropped and shrunk to 50x50 pixels to guarantee uniformity.

. After that, the data is saved and serialised. To preserve the records of earlier users, the new data is appended to data files that already exist**.**

2. Module for Recognition and Attendance (test.py) For general users, this is the primary execution module.

. After loading the training set, it fits the KNN classifier (knn.fit(FACES, LABELS)).

. It reads frames from the video feed constantly. When a face is identified, cv2.putText is used to display the name on the screen.

. User Interaction: The system initiates the attendance mechanism when the user hits the "o" key.

. Win32com.client is used. To deliver audio feedback ("Attendance Taken.."), dispatch ("SAPI.SpVoice").

. It looks for a CSV file that matches the current date. The user's name and the current timestamp are appended if the file already exists; otherwise, a new file with the headers ['NAME', 'TIME'] is created.

. Additionally, the module makes use of a backdrop picture (background.png) to give the camera stream a graphical overlay.

3. Module for Admin Dashboard (app.py) The organization's monitoring tool is this module.

. It makes sure that new attendance entries show up on the dashboard almost instantaneously by using st_autorefresh(interval=2000) to refresh the script every two seconds.

. It retrieves the particular CSV file for the current date using pd.read_csv().

. St.dataframe() is used to display the data, and style.highlight_max is used to make the records easy to see.

Data Flow

1) Enrollment: User Input (Name) + Camera Feed -> add_faces.py -> Update .pkl files.

2) Verification: Recognised Name -> Camera Feed -> test.py (KNN Prediction).

3) Logging: Press the "O" key -> test.py -> Attendance_DD-MM-YYYY.csv can be updated or created.

4) Observation: Attendance_DD-MM-YYYY.csv -> app.py -> Streamlit Web Interface.

VII. PROPOSED WORK

The three interconnected modules of the Smart Attendance System simplify the attendance procedure: **1.** Enrolment (Face Registration) The administrator executes add_faces.py and enters the user's name to register a user. Using OpenCV, the system records 100 facial frames. Before being saved as faces_data.pkl, each frame is flattened and shrunk to 50x50 pixels. This guarantees a varied dataset for precise training.

2. Instantaneous Identification and Marking Live operation is handled by the test.py module:

. Detection: Faces on the video feed superimposed on a personalised background (background.png) are identified using Haar Cascade Classifiers.

. Identification: Using the stored data, the KNN algorithm forecasts the user's identity.

. Marking: When the 'o' key is pressed, the system adds the name and timestamp to a daily CSV file (such as Attendance_27-11-2025.csv) and uses win32com to offer audible feedback ("Attendance Taken").

3. Dashboard for Administrators Using Streamlit, the app.py module offers a real-time monitoring interface. Every two seconds, it reads the daily CSV file and displays the data in a highlighted dataframe for immediate verification using st_autorefresh.

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VIII. RESULTS AND DISCUSSION

The accuracy and speed of the system were evaluated. Among the main conclusions are:

Measures of Performance

.Enrolment: It takes ten to fifteen seconds to collect one hundred face samples.

.Response Time: When a key is pressed, real-time recognition and audible feedback happen right away.

.Data management: To avoid data loss, daily CSV files are created automatically.

Comparison with Manual System

Feature	Manual System	Proposed Smart System
Method	Roll Call / Paper	Facial Biometrics (KNN)
Accuracy	Low (Proxy potential)	High (Visual Verification)
Time Taken	10-15 Mins/Class	< 5 Seconds/User
Monitoring	Static Registers	Real-Time Web Dashboard

The system successfully eliminates "Buddy Punching" and reduces the administrative burden of manual data entry.

IX. DISCUSSION

A solid and lightweight system for tracking attendance was made possible by the merging of Scikit-learn and OpenCV. Using Streamlit made it possible to quickly create an intuitive dashboard without the need for complicated web frameworks. For the best Haar Cascade detection, the system presently depends on steady lighting. Deep learning models, such as CNNs, may help future versions perform better in low light.

X. FUTURE SCOPE

The proposed Smart Attendance System demonstrates significant improvements in handling daily logs, automating the verification process, and monitoring student attendance data. However, there are many opportunities to further enhance the system, making it more intelligent, efficient, and adaptable. This system's potential range includes:

Creation of Mobile Applications Teachers and administrators will be able to utilise the system on portable devices by developing a feature-rich mobile application (for iOS and Android). This portability increases its dependability in a variety of settings by enabling attendance to be recorded in places

Integration of Deep Learning and Advanced AI Although K-Nearest Neighbours (KNN) is used in the existing system, AI can be further integrated to provide intelligent characteristics that significantly increase recognition accuracy. In order to manage complicated differences in illumination, facial angles, and accessories (such as masks or glasses), Convolutional Neural Networks (CNNs) must be used. Additionally, predictive analytics that predict student irregularity based on past attendance trends must be provided.

Advanced Documentation and Analysis Administrators may assess class participation, track attendance patterns in real time, and generate thorough reports for better decision-making by utilising advanced data analytics. Pie charts displaying presence percentages, the automatic creation of monthly PDF reports, and the identification of frequent absentees are all possible future additions.

Enhanced Liveness Detection and Security By implementing "Liveness Detection" capabilities, the system's security and integrity will be enhanced. In order to combat spoofing attacks, in which users attempt to register attendance using other people's photos or videos, algorithms that detect eye blinks or grins are used to determine whether the individual is physically present.

Connectivity to Notification Systems Easy communication with stakeholders is made possible by connecting the attendance system to email or SMS services. In order to improve outreach and guarantee prompt interaction regarding attendance difficulties, the system might automatically send "Low Attendance" warnings to students and "Absent" notices to parents or guardians.

Cloud-Based Synchronisation of Data Data accessibility is improved by moving the system from local CSV storage to a fully cloud-based architecture (like Firebase or AWS). This would make it possible for several classes to synchronise in real time, giving the principal or administrator rapid access to the entire institution's attendance data via a single dashboard.

Scalability in Big Organisations The system will be able to efficiently handle large-scale organisations with thousands of participants if the database architecture is improved. Even when the number of registered users increases dramatically, recognition speed will remain instantaneous if the face encoding retrieval procedure is optimised.

With these future enhancements, the Attendance System may become a comprehensive, intelligent, and flexible solution that effectively manages all verification requirements while providing administrators and users with a seamless experience.

XI. CONCLUSION

The procedure of documenting the presence of students or employees is effectively automated by the established Smart Attendance System. The solution removes the inefficiencies of human roll calls by utilising Python, OpenCV, and Machine Learning. A smooth user experience is guaranteed by the innovative combination of acoustic feedback and a real-time Streamlit dashboard. The system is a practical answer for contemporary business and educational settings since it is affordable, adaptable, and offers safe, unchangeable data.

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